PPP-B-41b
September 9, 1964

SUPERSEDING
Int. Fed. Spec. PPP-B-0041a (Army-GL
March 9, 1964 and
Fed. Spec. PPP-B-41
February 27, 1953

FEDERAL SPECIFICATION

BARRELS; WOOD, SLACK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** Slack wood barrels covered by this specification are suitable for shipping dry powderous materials or solids in bulk. Suitable bag-type barrel liners shall be used when necessary.

1.2 **Classification.**

1.2.1 *Types.* Slack barrels covered by this specification shall be furnished in one of the following types, as specified:

Type I—Light Duty (see 3.3.2, 6.2.1).

Type II—Heavy Duty (see 3.3.3, 6.2.2).

1.2.2 *Grade.* Slack barrels covered by this specification shall be of but one grade:

Grade A—Millrun.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

FF-N-103—Nails (Small) and Tacks; Cut.

FF-N-105—Nails, Wire, Brads and Staples.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly su plements as issued, is for sale on a subscripti basis by the Superintendent of Documents, U. Government Printing Office, Washington, D. 20402.

(Single copies of this specification and oth product specifications required by activities outsi the Federal Government for bidding purposes a available without charge at the General Servic Administration Regional Offices in Boston, Ne York, Washington, D.C., Atlanta, Chicago, Kans City, Mo., Dallas, Denver, San Francisco, L Angeles, and Seattle, Wash.

(Federal Government activities may obtain copi of Federal Specifications, Standards, and Han books and the Index of Federal Specifications a Standards from established distribution points their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipme and Storage.

MIL-STD-731—Quality of Wood Mer bers for Containers and Pallets.

(Copies of Military Specifications and Standar required by the contractor in connection with spe ific procurement functions, should be obtained fro the procuring agency or as directed by the contrac ing officer.)

3. REQUIREMENTS

3.1 **Materials.**

3.1.1 *Species of wood.*

3.1.1.1 *Group A.* Wood shall be ash, beec birch, Douglas fir, elm, gum, hackberry, m ple, oak, yellow pine, yellow poplar.

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3.1.1.2 *Group B.* Wood shall be basswood, box elder, cottonwood, spruce, sycamore, western hemlock.

3.1.1.3 *Wood defects.* Wood shall be free of knots within two inches of the end of each stave, heading, cleat, head liner or hoop; elsewhere sound knots smaller than 3/4-inch diameter shall be admitted. Wood shall be free of unsound knots, holes, slanting shakes longer than 1-1/2 inches, decay, splits, heavy stain, and cross grain (slope of grain) steeper than 1 inch in 10 inches.

3.1.1.4 *Moisture content.* The moisture content of staves, heading and cleats shall be not less than 7 percent nor more than 11 percent of their dry weight at time of manufacture of the barrels. The moisture content of head liners and wood hoops shall be not less than 12 percent nor more than 19 percent at time of manufacture of the barrels (see 4.2.2).

3.1.1.5 *Millrun barrel construction.* Millrun heading, shall be made of group A or B woods, in accordance with table I.

3.1.2 *Steel.* Steel for steel cleats and steel hoops shall be mild steel, hot rolled cooperage grade hoop steel.

3.1.3 *Fasteners.*

3.1.3.1 *Nails, wire, brads, and staples.* Nails and staples shall conform to requirements of FF-N-105.

3.1.3.2 *Nails, (small) and tacks, cut.* Small nails and tacks shall conform to FF-N-103.

3.1.3.3 *Rivets.* Rivets shall be standard cooper's rivets.

3.2 Fabrication.

3.2.1 *Staves.* Staves shall be tongued and grooved or butt-jointed, and wood used for staves shall conform to requirements of 3.1.1.1, 3.1.1.3 and 3.1.1.4. The thickness of staves shall be as specified in table I for the size and capacity of barrel specified.

3.2.2 *Heads.* Heads shall be fabricated as specified in 3.2.2.1 and 3.2.2.2. The minimum thickness of the heads, and the maximum number of cants in any head, shall be as specified in table I, and the minimum width of each cant shall be 2-1/2 inches. Wood shall conform to requirements in 3.1.1.

3.2.2.1 *One-piece heads.* Heads shall be considered one piece if the cants are tongued and grooved and glued, butt-jointed and glued, or butt-jointed and fastened with not less than two corrugated fasteners per joint. Corrugated fasteners shall be driven on the inner face and shall not penetrate through the head to the outer face. Heads shall be surfaced on a minimum of one side and shall be turned to size with a standard head bevel.

3.2.2.2 *Cant heads.* Cant heads shall be made of separate cants which are butt joined. Heads shall be surfaced on a minimum of one side and shall be turned to size with a standard head bevel.

3.2.3 *Cleats.*

3.2.3.1 *Wood cleats.* Wood cleats shall be fabricated from material conforming to 3.1.1. Cleats shall be 1/2 inch thick and shall be shaped and beveled to fit the chime. Width dimensions of cleats for type I and type II barrels shall be as specified in table II. The tolerance for width dimensions shall be ± 1/4 inch.

3.2.3.2 *Steel cleats.* Steel cleats shall be fabricated from material conforming to 3.1.2. Cleats shall be not less than 1-7/8 inch wide and 0.0625 inch (16 gage) thick, and shall have a central longitudinal bead not shorter than 1 inch less than the diameter of the head. Each cleat shall have a minimum of five holes for the application of fasteners. The cleat ends shall be formed to fit over the chime and under the head hoop, and the cleat shall extend a minimum of 7/16 inch beyond the head hoop. Steel cleats shall not be specified for barrels with heads larger than 19 1/8 inch diameter.

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TABLE I.—*Millrun barrel construction*

Net weight of contents	Diameter	Heading		Staves		Hoops[5]				
						Metal hoops				Wood hoops
						Head		Bilge[2] and quarter[3]		
		Thickness (nominal)[1] Wood groups A or B	No. of cants, maximum	Length	Thickness, nominal[1]	Width	Thickness	Width	Thickness	
Pounds	*Inches*	*Inch*		*Inches*	*Inch*	*Inches*	*Gage*[4]	*Inches*	*Gage*	*No.*
To 450	13	½	3	21 24	⅝	1-⅝	22	1⅜	23	6
451-600	13	⅝	3	21 24	⅝	1-⅝	22	1⅜	23	6
601-1000	13	⅝	3	21 24	½	1-⅝	21	1⅜	23	8
To 450	14	½	3	21 24	⅝	1-⅝	22	1⅜	23	8
451-600	14	⅝	3	21 24	⅝	1-⅝	22	1⅜	23	8
601-1000	14	⅝	3	21 24	½	1-⅝	21	1⅜	23	10
To 450	15	½	3	21 24 28½ 30	⅝	1-⅝	22	1⅜	23	8
451-600	15	⅝	3	21 24 28½ 30	⅝	1-⅝	22	1⅜	23	8
601-1000	15	⅝	3	21 24 28½ 30	½	1-⅝	21	1⅜	23	10
To 450	16	½	3	24 28½ 30	⅝	1-⅝	22	1⅜	23	8
451-600	16	⅝	3	24 28½ 30	⅝	1⅝	22	1⅜	23	10
601-1000	16	⅝	3	24 28½ 30	½	1⅝	22	1⅜	23	12
To 450	17⅛	½	4	24 28½ 30	⅝	1⅝	22	1⅜	23	8

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TABLE I.—*Millrun barrel construction (cont'd)*

Net weight of contents	Diameter	Heading: Thickness (nominal)[1] Wood groups A or B	Heading: No. of cants, maximum	Staves: Length	Staves: Thickness, nominal[1]	Hoops[5] Metal hoops: Head: Width	Head: Thickness	Bilge[2] and quarter[3]: Width	Bilge[2] and quarter[3]: Thickness	Wood hoops
Pounds	*Inches*	*Inch*		*Inches*	*Inch*	*Inches*	*Gage*[4]	*Inches*	*Gage*	*No.*
451-600	17⅛	⅝	4	24 28½ 30	⅝	1⅝	22	1⅝	23	10
601-1000	17⅛	⅝	4	24 28½ 30	½	1⅝	21	1⅝	23	12
To 450	19⅛	½	5	30	⅝	1⅝	22	1⅝	23	8
451-600	19⅛	⅝	5	30	⅝	1⅝	22	1⅝	23	10
601-1000	19⅛	⅝	5	30	½	1⅝	21	1⅝	23	12
To 450	21	½	5	30 34	⅝	1⅝	21	1⅝	23	8
451-600	21	⅝	5	30 34	⅝	1⅝	21	1⅝	23	12
601-1000	21	⅝	5	30 34	½	1⅝	20	1⅝	23	12

[1]Thickness of heading and staves are nominal as stated in the table. Actual thicknesses in the finished barrel may be as much as 1/16 inch scant of the tabuler thickness.

[2]Wire bilge hoops may pe used on barrels when the net weight of contents does not exceed 250 pounds.

[3]Quarter hoops not required when the net weight of contents does not exceed 450 pounds.

[4]Gage refers to Birmingham wire gage.

Gage	Inches
20	0.035
21	.032
22	.028
23	.025

[5]Applies to flat hoops. Beaded head hoops shall be 1⅝ inches and beaded bilge and quarter hoops 1 inch in width.

3.2.4 *Head liners.* Head liners shall have a 3/16-inch by 1/2-inch cross section and shall be of sufficient length to support the ends of the head cants. Head liners shall be of elm and shall meet requirements of 3.1.1.3 and 3.1.1.4. Liners shall be formed to fit the curve of the chime.

3.2.5 *Hoops.*

3.2.5.1 *Wood hoops.* The minimum sectional dimensions of wood hoops shall be 9/32 by 5/32 by 1-3/8 inches. One end of each hoop shall be tapered edgewise. The other end shall be tapered flatwise. The ends shall be joined with not less than four 13/16-inch clout nails or 18-ounce cooper's tacks, staggered, three 0.072 inch (15 gage) staples, or any combination thereof. The hoops shall be constructed of elm wood and conform to requirements of 3.1.1.3 and 3.1.1.4. The hoops shall be formed to fit snugly when driven into place.

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3.2.5.2 *Metal hoops.* Metal hoops shall be constructed of material conforming to 3.1.2. The minimum width of metal hoops, either flat or after beading or rolling of an edge, and thickness of metal hoops shall be as stipulated in table I. The ends of each hoop shall be joined with two rivets 0.192 inch (6 gage) diameter. Rivets shall be located at the center of the width of the hoop, not less than 5/8 inch from the ends and spaced not less than 1 inch. They shall be flush with the inside of the hoop and the outside head shall be flat. Hoop shall be flared and formed to fit snugly when driven into place.

3.2.5.3 *Wire hoops.* Wire hoops shall be constructed of steel wire which shall conform to minimum requirements of 3.1.2 and minimum wire thickness shall be 0.185 inch (10 gage) diameter. The ends of each hoop shall be butt welded or joined with a twisted splice, at the option of the contractor. Wire shall be formed to fit snugly when driven into place.

3.2.6 *Steel straps.* When specified, steel straps shall be not less than 7/8 inch wide and 0.032 inch (21 gage) thick by not less than 6 inches long. Each end shall be punched in not less than 4 places so as to form prong fasteners not less than 3/8 inch long.

3.3 Assembly. (See figure 1 for illustration).

3.3.1 *General.* The general assembly requirements shall apply to types I and II barrels.

3.3.1.1 *Staves.* Staves, conforming to 3.2.1, of varying widths, shall be assembled so that the bilge and chime are round and barrels of the same nominal size have approximately the same capacity.

3.3.1.2 *Croze.* The depth of the croze shall be not less than 1/8 inch nor more than 5/32 inch. The croze shall conform to the bevel of the head.

3.3.1.3 *Chime.* The top of the head sha be not less than 1/2 inch nor more than 5/ inch from the ends of the staves.

3.3.1.4 *Bilge and quarter hoops.* The bilg and quarter hoops shall be placed aroun the assembled staves and driven level an tight. The bilge and quarter hoops shall b secured at not less than 4 points at approxi mately equal spacing with nails, hoop faster er nails, or staples. If points of fastening extend through the staves, they shall b clinched.

3.3.1.5 *Head hoops.* The head hoops sha be put around the assembled staves and th heads put in place. The head hoops shall b driven into place level and tight. Head hoop shall be secured with 0.106-inch (12 gage by 1-inch (1-1/4-inches for use with woo hoops) nails, 0.121-inch (11 gage) hoop sta ples spaced approximately 6 inches, or nc less than ten 0.048-inch (18 gage) wir stitches. Fastenings shall pass through hoo and staves into the head so that each hea cant shall have not less than two faster ings.

3.3.2 *Type I barrels.* Type I barrels sha be assembled using wood hoops conformin to 3.2.1.5.1; or metal hoops conforming t 3.2.5.2; or wire hoops conforming to 3.2.5.3 Wire hoops shall be used as bilge hoops onl and shall fit snugly when driven into plac (see note 2, table I). Heads shall conform t 3.2.2.

3.3.2.1 *Barrel reinforcement.* When spec fied, type I barrels shall be reinforced b one or a combination of several of the fo lowing methods.

3.3.2.1.1 *Wood cleats.* Wood cleats, cor forming to 3.2.3.1, shall be applied as spec fied in table II, outside of and across th grain of the head and fastened with not les than 8 nails 0.080-inch (14 gage), diamete not shorter than 1/8 inch less than the com bined thickness of head and cleat. The nail shall be staggered in two rows and if the extend through the head, shall be clinched

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They shall be spaced so that each head cant shall have at least two nails. Two nails or staples shall be driven through hoop and staves into each end of a cleat. Nails shall be 0.106 inch (12 gage) by 1-1/8 inches (1-3/8 inches for use with wood hoops). Staples shall be 0.121 inch (11 gage).

TABLE II.—*Wood cleats*

Net weight of barrel contents	Barrel diameter	Width of cleat	
		Type I, barrel	Type II, barrel
Pounds	*Inches*	*Inches*	*Inches*
Up to 450	13	—	4
450 to 600	13	—	4
600 to 1000	13	—	5
Up to 450	14	—	4
450 to 600	14	—	5
600 to 1000	14	—	5
Up to 450	15	—	4
450 to 600	15	—	5
600 to 1000	15	4	5
Up to 450	16	—	4
450 to 600	16	—	5
600 to 1000	16	4	6
Up to 450	17⅛	—	4
450 to 600	17⅛	4	5
600 to 1000	17⅛	5	6
Up to 450	19⅛	4	5
450 to 600	19⅛	5	5
600 to 1000	19⅛	5	6
Up to 450	21	4	5
450 to 600	21	5	5
600 to 1000	21	5	6

3.3.2.1.2 *Steel cleats.* Steel cleats, conforming to 3.2.3.2, shall be applied on the outside of and across the grain of the head and fastened with not less than five 4-penny box nails driven through cleat and head and clinched. They shall be staggered. Cleat ends shall be bent back on the head hoop. Steel cleats shall not be specified for barrels with heads larger than 19-1/8 inch diameter.

3.3.2.1.3 *Head liners.* Two head liners, conforming to 3.2.4, shall be applied across the grain of each head on opposite sides of the circumference. Each head liner shall cover approximately one-fourth of the circumference and shall be fastened to the chime with not less than four staples or 2-penny common nails spaced approximately five inches. Head liners shall be applied to all barrels with wooden hoops in which the weight of the contents exceeds 250 pounds.

3.3.2.1.4 *Strapping.* One strap, conforming to 3.2.6, shall be applied to each end of a wood cleat by driving the prongs into the cleat. The strap shall then be bent over the chime and head hoop and the prongs driven into the stave.

3.3.3 *Type II barrels.* Type II barrels shall be assembled using metal hoops conforming to 3.2.5.2, and heads shall conform to 3.2.2.1.

3.3.3.1 *Barrel reinforcement.* Type II barrels shall be reinforced as outlined in 3.3.3.1.1, 3.3.3.1.2 and 3.3.3.1.3.

3.3.3.1.1 *Wood cleats.* Wood cleats shall conform to 3.3.2.1.1.

3.3.3.1.2 *Head liners.* Head liners shall conform to 3.3.2.1.3.

3.3.3.1.3 *Strapping.* Strapping shall conform to 3.3.2.1.4.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for inspection.** Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

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4.2.1 *Component and material inspection.* In accordance with 4.1 the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2.2 *Intermediate testing of components and materials.* Tests shall be performed on representative samples selected from each lot or batch of stave, heading, cleats, head liners, or hoops used in the fabrication of the end item, to determine moisture content as specified in 3.1.1.4. Testing shall be performed in accordance with the requirements of MIL-STD-731. The sample unit shall be one stave, heading, cleat, head liner, or hoop, and the sample size shall be five. When tested by the moisture meter method, the average of 3 determinations taken on each component shall represent the moisture content for the component. The average of all tests performed shall not be more or less than the specified limits.

4.3 **Examination of the end item.** Defects found during examination shall be in accordance with table III. The sample unit shall be one barrel and lot size shall be expressed in number of the sample units. The sample size shall be in accordance with inspection level S-2 and the acceptable quality level (AQL's) expressed in terms of defects per hundred units (DHU) shall be 6.0.

TABLE III.—*Examination of the end item*

Examine	Defect
Capacity (net weight)	Not as specified in table I
Wood	Not material specified
	Knot within two inches of end of piece
	Sound knot greater than ¾ inch diameter
	Unsound knots, holes, decay, splits, or stains
	Slanting shake greater than 1½ inches
	Cross grain greater than 1 inch in 10 inches

TABLE III.—*Examination of the end item (cont'd)*

Examine	Defect
Staves:	
Dimensions	Not within tolerance indicated (i table I)
Croze	More than 3/16 inch or less th ⅛ inch deep
Chime	More than ⅝ inch or less than inch from end of stave.
Head:	
Cants	Number of cants in any head mo than specified (see table I)
	Any cant less than 2½ inch width
	Not within tolerance indicated (s table I)
	Not smooth surface on one side
	Not turned to standard head be
Cleats	Not positioned outside and acr head grain (see table II)
	Not dimensions specified
	Steel cleats do not pass under he hoops
	Steel cleats not bent back over he hoops
	Steel cleats not beaded as specifi
	Wooden cleats not chamfered to chime
	Wooden cleats not fastened to hea as specified
	Nails, staples not size specified
	Nails and staples not applied specified
	Less than specified number
Head liners	Not material specified (see 3.2.4)
	Not dimensions specified (see 3.2.
	Not applied and secured as specifi
Hoops:	
Material	Not material specified
Dimensions	More or less than specified in tal I
Construction	Ends not joined as specified
	Not flared and formed to fit snug as specified
	Rivets (when specified) wro size, not located on joint, not fl head and flush inside hoop specified
	Cracks, tears, sharp edges
	Not secured as specified
	Not tight and level as specifi (see 3.3.1.4 and 3.3.1.5)
	Not beaded or rolled edge as spe fied
	Wire hoops used for other th bilge hoops

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TABLE III.—*Examination of the end item (cont'd)*

Construction (cont'd)	Fastenings piercing staves, not clinched
	Less than specified number per barrel (see table I)
Strap	Not material specified
	Less than minimum specified dimensions
	Punched in less than four places
	Any prong less than ⅝ inch long
Marking	Missing, illegible, incorrect, incomplete
	Not located as specified

4.4 Standby test samples. The Government reserves the right to withdraw and hold standby test samples of components, or finished product, or both, (quantity of which shall be not more than twice that specified for this specification) for inspection purposes. These samples if not used, will be returned to the vendor.

4.5 Tests. Test procedures (see 4.2.1) and controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide as a minimum, such quality assurance, the supplier shall use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein shall govern.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Not applicable.

5.2 Packing. Unless otherwise specified in the contract or order, packing shall be level C.

5.2.1 *Level C.* Empty barrels, headings and cleats shall be shipped in a manner which is acceptable by common carrier for safe transportation to point of destination at the lowest transportation rate for such supplies.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special markings required by the contractor's order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Slack barrels. A slack barrel is a round, bulging, coopered, wooden container with sides consisting of staves and ends of round, flat heads, the staves and the heads being held together with hoops. It is so constructed that its joints are not necessarily tight to the passage of liquids. Barrels with staves shorter than 24 inches are commonly known as kegs.

6.1.1 *Sizes.* The barrels described herein will provide sound, substantial containers for many commodities and they may be obtained in a considerable range of sizes, the selection of which depends upon the commodities to be shipped and to a large extent on custom. Probably the most common size of slack barrel has 19-1/8-inch heads 30-inch staves. Other sizes and qualities of slack barrels may be obtained. These, however, are not covered in this specification and arrangements for their purchase should be made only after obtaining the advice of skilled coopers.

6.2 Intended use. Slack barrels covered by this specification are intended for use by the Government or for the shipment of commodities purchased by the Government when prescribed in the applicable commodity specification.

6.2.1 *Type I.* Type I barrels covered by this specification are primarily intended for normal shipping, handling and storage requirements, and where the additional strength of the type II barrel is not required.

6.2.2 *Type II.* Type II barrels covered by this specification are primarily intended for rigorous shipment, repeated handling, protected and unprotected storage and possible extreme climatic conditions are contemplated. Type II barrels meet all requirements for overseas shipments.

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6.3 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, grade, and quantity of barrels desired.
(*c*) Barrel capacity, head diameter, and stave length.
(*d*) Reinforcement required for type I barrels.

6.4 **Definitions of slack barrel terms.**

6.4.1 *Bilge.* The bilge is the largest circumference of a barrel.

6.4.2 *Cant.* A cant is any board forming a part of a head.

6.4.3 *Chime.* The chime is that portion of the staves extending beyond the outside surface of the head.

6.4.4 *Cleat.* A cleat (or batten) is a board of specified dimensions placed across the head of a barrel and fastened to the barrel for reinforcement purposes.

6.4.5 *Croze.* The croze is a V-shaped groove cut into the inside circumference of the barrel near the ends of the staves to receive the beveled edge of the head.

6.4.6 *Decay.* Disintegration of wood substance due to action of wood-destroying fungi; also known as dote and rot.

6.4.7 *Head.* The head is a formed, circular member that fits into the croze to form the ends (top and bottom) of barrel.

6.4.8 *Heavy stain.* Difference in color so pronounced as practically to obscure the grain of the wood.

6.4.9 *Hoop.* A hoop is an annular ring of steel, wire, or wood fastened to the staves around the circumference of the barrel to hold the barrel together. The head hoops are those at the end; the bilge hoops are those nearest the bilge; the quarter hoops are those near the ends, between the head hoops and the bilge hoops.

6.4.10 *Slanting shake.* A separation along the grain extending through the piece at an angle of less than 45°.

6.4.11 *Slight roughness.* Grain torn in cutting staves, not more than 1/16 inch deep.

6.4.12 *Stave.* A stave is an arched, vertical member used to form the sides of a barrel.

6.5 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail:

Barrels, wooden, not otherwise indexed by name, slack not nested.
Carload minimum weight, 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Barrels, wooden, not otherwise indexed, slack not nested.
Truckload minimum weight, 10,000, 12,000, 20,000, or 30,000 pounds, subject to Rules 115 and 140, National Motor Freight Classification.

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APPENDIX 1

REQUIREMENTS FOR OPENING, FILLING, AND CLOSURE OF EMPTY SLACK BARRELS AND APPLICATION OF BAG LINERS

10. REQUIREMENTS FOR OPENING, FILLING, AND CLOSURE.

10.1 General. Don't knock a head in with an axe or hammer. If recoopering by the packer is necessary, take out the head with the least number of pieces if multiple piece head is used. After selecting the end of the empty barrel to be opened, turn it bottom side up and set the bottom head as specified in paragraph 3.3. Test the quarter and bilge hoops (next to bottom head) for tightness. If at all loose, don't hammer and pound, tap lightly and keep going around the barrel using suitable tools. These tools include a cooper's hammer, hoop driver, catch ring, chime maul and a hand stapling machine for stapling heads. Hand stapling machines drive staples accurately through the hoop and stave into the head. They are more accurate than hand nailing, faster and have less tendency to split heads. Staples may be extracted with a screwdriver, and the heads removed intact.

10.2 Opening. Release the top head hoop by tapping gently upward with a hammer until about half the hoop is above the edge of the chime. This will loosen the staves so that a light blow on one side of the head will drop it to the bottom of the barrel. The head then should be removed. Drive the hoop down again level with the chime. If a hickory catch ring is available, it can be driven down below the head hoop to take the place of the head hoop, which can be removed entirely. Do not allow a barrel to stand for any length of time with head hoops removed. Barrels with top head removed will collect dust and dirt.

10.3 Filling. Slack barrels shall be loaded to full capacity by agitation when applicable during filling operation to insure a minimum shifting of the contents in transit. Suitable filler material should be used to assure complete filling of the barrel.

10.4 Closing. (If a bag liner is used, see par. 20). After the barrel is filled and the bag liner (if any) tied, replace the head, taking care that the liner is clear of the croze so that the head will not pinch or puncture the liner. Drive the head hoop upward about half its width, to release the staves. Place one edge of the head in croze and press or tap down lightly with a hammer until the head is seated in croze all around the barrel. As the head slides into place, keep tapping top hoop with hammer gently around the barrel. Avoid driving hoop down tightly at any one place, but endeavor to bring it down evenly all around the circumference of the barrel. When the head hoop is driven tight, be sure the head is firmly secured. The barrel is now ready to have the hoops and head nailed or stapled as specified in paragraph 3.3.

10.5 Reinforcement. Reinforcement shall be applied in accordance with the requirements of 3.3.2.1.

20. APPLICATION OF BAG LINERS.

20.1 Slack barrels may be provided with a separate bag liner to protect the contents from contamination, abrasion, to facilitate removal of contents, or to protect contents from moisture and sifting. The bag liner should be inserted with an inserting device known as a bag former. Place the bag former into the bag liner as far as it will go. Then fold the ears of the bottom of the bag liner inwardly and insert the bag former and the bag liner into the upright barrel. Leaving the bag former in the barrel, fold the ends of the liner that protrude loosely over the chime of the barrel. Do not fold the liner down tight against the top hoop, but leave a space of several inches between the fold of the liner and the chime of the barrel. The weight of the material used to fill the barrel will draw the liner down. The bag form

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er is now removed from the barrel, making sure that the liner lies smooth in the bottom of the barrel. The barrel is now ready to fill. After filling gather the liner together over the contents and force out the air. Tie the liner together as close to the top of the ear as possible. If the liner is tied too close to the contents, the liner will be broken when the contents shift. The barrel is now ready to be headed up as described in paragraph 10.

30. MARKING

30.1 Marking shall be in accordance with the requirements of the commodity specification.

40. INSPECTION PROCEDURES

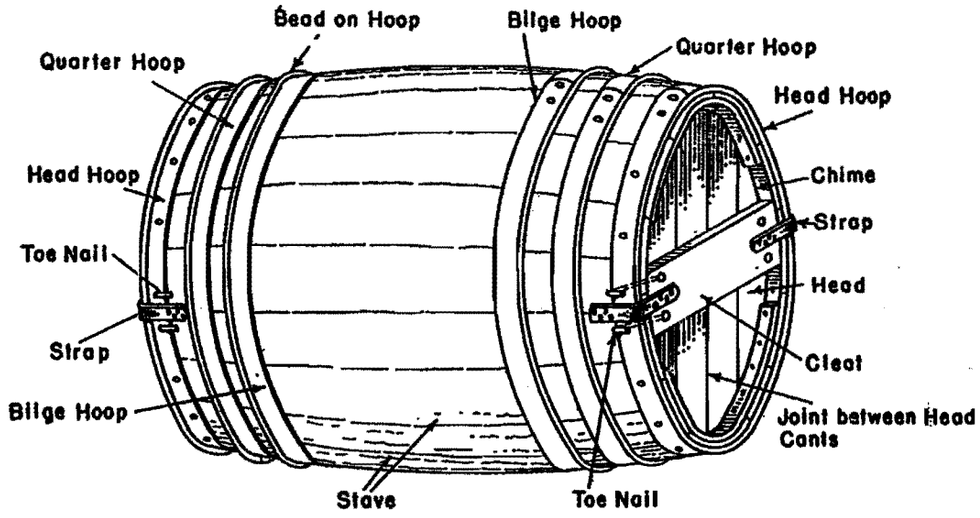
40.1 Inspection. Barrels shall be examined to determine compliance with the requirements of this appendix. Sampling shall be conducted in accordance with the provisions of MIL-STD-105.

40.1.1 *Examination for filling, closing and marking.* Classification of defects for filling, closure, and marking shall be as indicated in table IV. The sample unit shall be one filled and closed barrel. The lot size for determining the sample size shall be expressed in barrels. The inspection level shall be S-3. The acceptable quality level (AQL), expressed in defects per hundred units, shall be 2.5 for major defects and 10.0 for total defects.

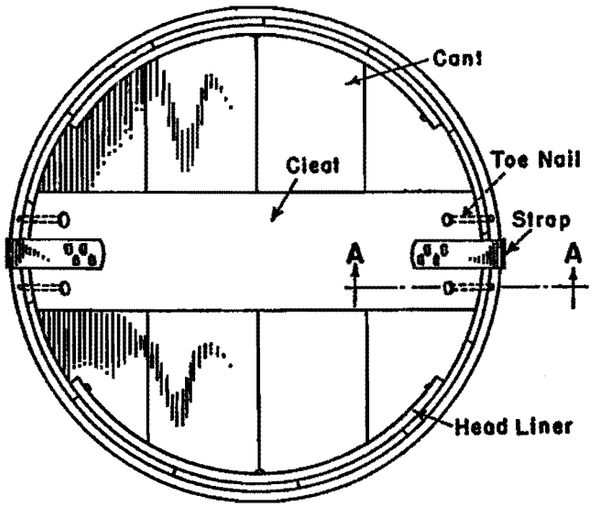
TABLE IV.—*Examination of filled barrels*

Examine	Defect	Major	Minor
Condition of barrel	Broken or fractured wooden component	X	
Filling	Not loaded to full capacity		X
	Filler material missing when applicable to assure complete fill		X
Bag liner (when applicable)	Bag liner missing	X	
	Bag liner not arranged or closed as specified		X
	Bag liner pinched or punctured	X	
Closure	Head not seated in croze entirely around the barrel	X	
	Head not firmly secured in place	X	
	Top hoop not driven tight or evenly around circumference of barrel		X
Marking	Item nomenclature missing, incorrect or illegible	X	
	Marked net weight does not conform to weight limitations of the applicable specification		X
	Other information missing, incorrect or illegible.		X

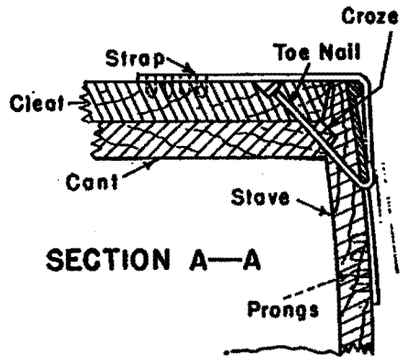
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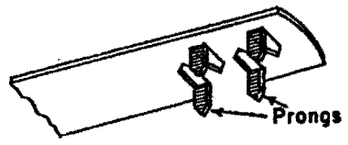
PERSPECTIVE



TOP PLAN



SECTION A—A



STRAP

FIGURE 1

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